

Krrish Baghla

Project: MMRDR (Multimodal Retinal Disease Recognition)

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Research Intern — IIT BHU

WORKFLOW & ENGINEERING

- **Data Pipeline Engineering:** Automated clinical data extraction via Figshare API, followed by file merging and systematic extraction protocols.
- **Multimodal Organization:** Categorized the dataset into three distinct modalities—**CFP** (Color Fundus Photography), **OCT**, and **UWF**—and established dedicated directory structures for each.
- **Dataset Preparation:** Implemented a rigorous 80% training and 20% testing split across 11,118 images to ensure robust validation.
- **Resolution Management:** Standardized all inputs at 512x512 resolution to maintain high-fidelity features for diagnostic lesion identification.

MODEL ITERATIONS & MILESTONES

- **ResNet-50 (Baseline Grading):** Targeted an accuracy benchmark of 0.823. Achieved a peak **Grading Accuracy of 0.8107** (98.5% of target) after three training cycles for 5-grade (0–4) DR classification.
- **RETFound Architecture:** Evaluated the state-of-the-art RETFound model; determined that compute requirements and epoch counts exceeded available hardware resources for optimal results.
- **Multitask ResNet-50:** Developed a secondary ResNet-50 architecture trained to simultaneously predict DR grades (0-4) and perform lesion detection.
- **Spatial Detector:** Engineered a dedicated localization detector capable of identifying the precise spatial coordinates of lesions within the retinal scans.

TECHNICAL LEARNINGS

- **Checkpointing Logic:** Architected logic to automatically save the "best model" during training cycles to prevent progress loss and capture optimal weights.
- **Regularization Techniques:** Applied **L2 Regularization (Weight Decay)** to mitigate overfitting and improve generalization.
- **Pipeline Mastery:** Developed custom PyTorch Train/Test DataLoaders for efficient high-resolution image streaming.
- **Deep Learning Toolkit:** Gained proficiency in **CrossEntropyLoss**, **Adam Optimizer**, **CosineAnnealingLR** for scheduled learning rates, and **GradScaler** for Mixed Precision training.

- **Annotation Theory:** Distinguished critical differences between **Bounding Box** localization and **Pixel-Level** (Segmentation-based) annotation for medical diagnostics.

CURRENT STATUS & CHALLENGES

- **Overfitting:** The model currently exhibits signs of overfitting despite initialization of regularization; further tuning of hyperparameters is required.
- **Dataset Imbalance:** Identified significant class imbalance within the 0-4 DR grades, which is impacting the sensitivity of higher-grade classifications.